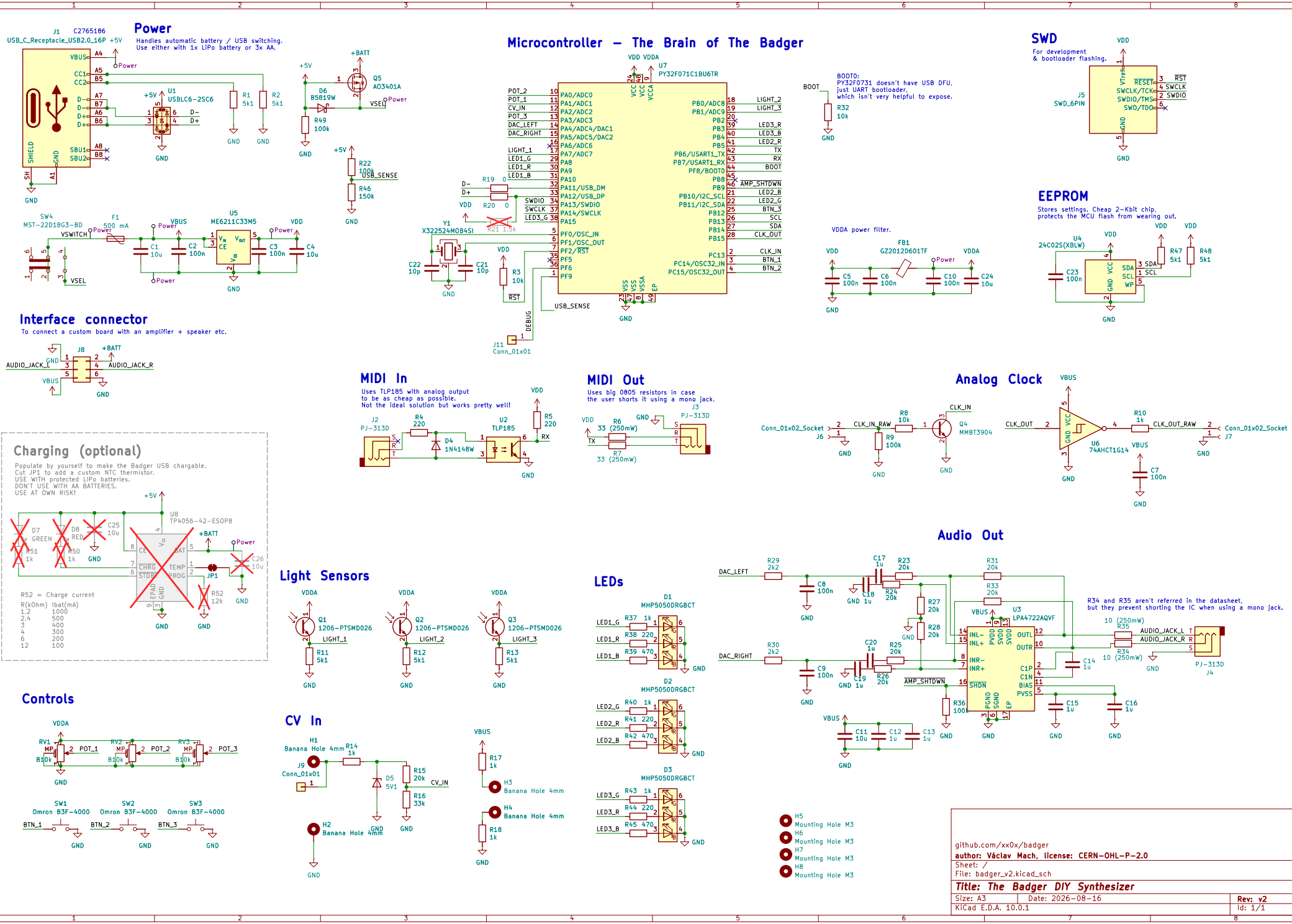

Power
Handles automatic battery / USB switching.
Use either with 1x LiPo battery or 3x AA.
Microcontroller – The Brain of The Badger
SWD
For development
& bootloader flashing.
EEPROM
Stores settings. Cheap 2-Kbit chip,
protects the MCU flash from wearing out.
Interface connector
To connect a custom board with an amplifier + speaker etc.
MIDI In
Uses TLP185 with analog output
to be as cheap as possible.
Not the ideal solution but works pretty well!
MIDI Out
Uses big 0805 resistors in case
the user shorts it using a mono jack.
Analog Clock
Charging (optional)
Populate by yourself to make the Badger USB chargable.
Cut JP1 to add a custom NTC thermistor.
USE WITH protected LiPo batteries.
DON'T USE WITH AA BATTERIES.
USE AT OWN RISK!
Light Sensors
LEDs
Audio Out
Controls
CV In
github.com/xx0x/badger
author: Václav Mach, license: CERN-OHL-P-2.0
Sheet: /
File: badger_v2.kicad_sch
Title: The Badger DIY Synthesizer
Size: A3
Date: 2026-08-16
Rev: v2
KiCad E.D.A. 10.0.1
Id: 1/1